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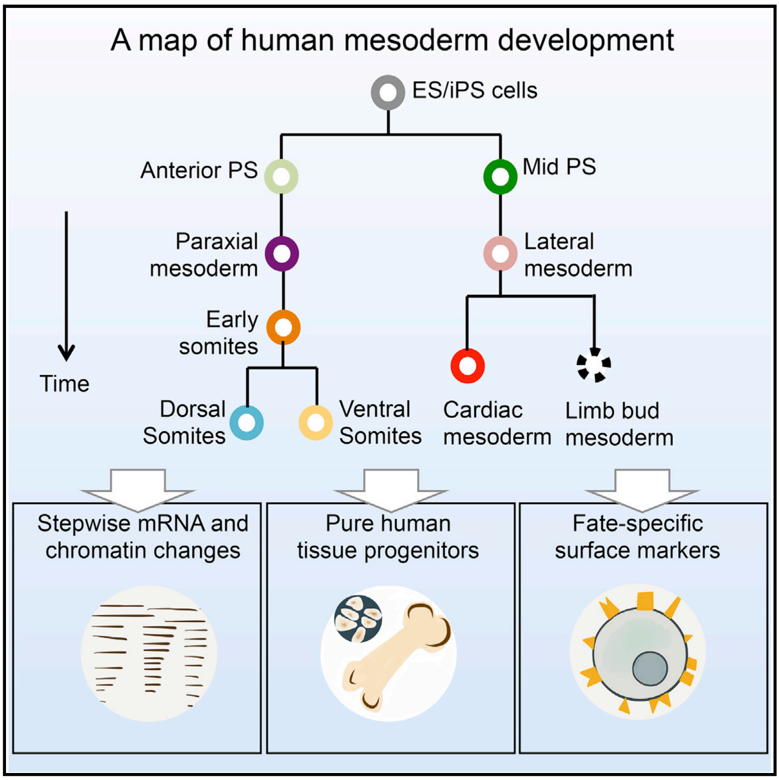
Mapping the Pairwise Choices Leading from Pluripotency to Human Bone, Heart, and Other Mesoderm Cell Types

Graphical Abstract



Highlights

- Stepwise map of competing signals guiding human mesoderm development
- Efficient human mesoderm induction by blocking formation of unwanted fates
- ESC-derived human bone progenitors and heart precursors engraft in vivo
- A transient segmentation program in human embryogenesis marked by *HOPX*

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In Brief

The lineage roadmap of human mesoderm development reveals transient developmental processes such as human somite segmentation and enables the generation and isolation of transplantable human bone and heart progenitors.

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Mapping the Pairwise Choices Leading from Pluripotency to Human Bone, Heart, and Other Mesoderm Cell Types

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SUMMARY

Stem-cell differentiation to desired lineages requires navigating alternating developmental paths that often lead to unwanted cell types. Hence, comprehensive developmental roadmaps are crucial to channel stem-cell differentiation toward desired fates. To this end, here, we map bifurcating lineage choices leading from pluripotency to 12 human mesodermal lineages, including bone, muscle, and heart. We defined the extrinsic signals controlling each binary lineage decision, enabling us to logically block differentiation toward unwanted fates and rapidly steer pluripotent stem cells toward 80%–99% pure human mesodermal lineages at most branchpoints. This strategy enabled the generation of human bone and heart progenitors that could engraft in respective in vivo models. Mapping stepwise chromatin and single-cell gene expression changes in mesoderm development uncovered somite segmentation, a previously unobservable human embryonic event transiently marked by *HOPX* expression. Collectively, this roadmap enables navigation of mesodermal development to produce transplantable human tissue progenitors and uncover developmental processes.

INTRODUCTION

Waddington's developmental landscape drawings (Waddington, 1940) depicted how differentiating stem cells negotiate a cascade of branching lineage choices, avoiding alternate fates at each juncture to decisively commit to a single lineage (Graf and Enver, 2009). To navigate this brachiating landscape and efficiently differentiate stem cells into desired cell-types for regenerative medicine, one must (1) catalog transitional lineage intermediates, (2) map the sequence of pairwise lineage choices through which such intermediates are formed, and (3) discover the positive and negative signals that specify or repress cell fate at each lineage branchpoint. Despite successes in charting lineage intermediates in mammalian tissues, key lineage branchpoints remain controversial and it has been impossible to systematically identify the extracellular signals that control cell fate at each branchpoint.

With the three above goals in mind, here we map the landscape of human mesoderm development in order to coherently guide stem-cell differentiation (Figure 1A). Mesoderm development begins with differentiation of pluripotent epiblast cells into the primitive streak, which then segregates into paraxial and lateral mesoderm, among other lineages (Lawson et al., 1991; Rosenquist, 1970; Tam and Beddington, 1987). Paraxial mesoderm segments into somites, which are fundamental building blocks of trunk tissue (Figure 1A, purple shading) (Pourquié, 2011). Somites are then patterned along the dorsal-ventral axis. The ventral somite (sclerotome) generates the bone and cartilage of the spine and ribs, whereas the dorsal somite (dermomyotome) yields brown fat, skeletal muscle, and dorsal dermis (Christ and Scaal, 2008). Separately, lateral mesoderm (Figure 1B, red shading) gives rise to limb bud mesoderm (Tanaka, 2013) and cardiac mesoderm (Später et al., 2014), the latter of which generates cardiomyocytes and other heart constituents. Various transcription factors (TFs) and signaling molecules regulating mesoderm development in model vertebrates have been identified, broadly outlining the developmental landscape (Kimelman, 2006; Schier and Talbot, 2005; Tam and Loebel, 2007).

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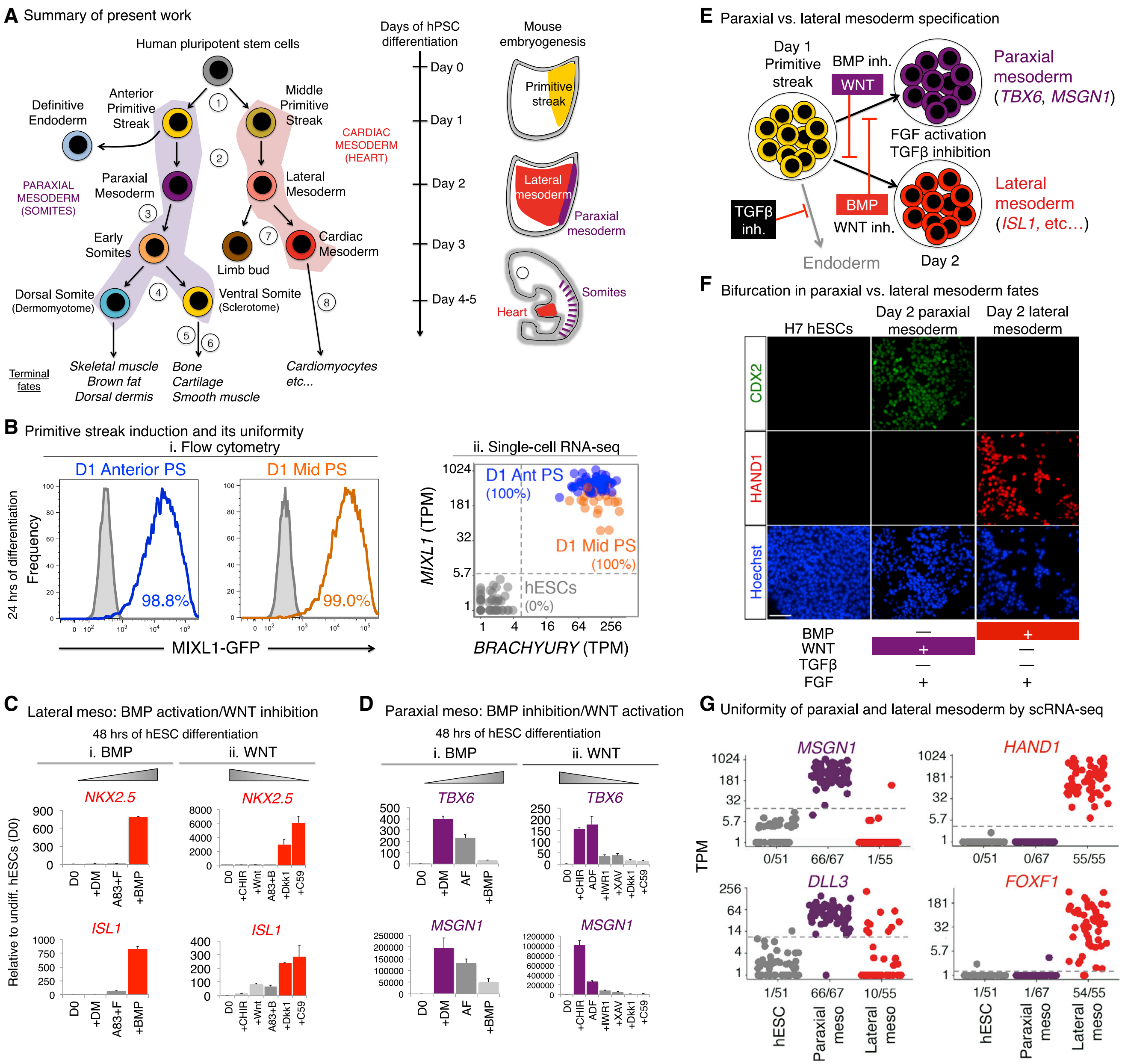

Figure 1. Formation of Human Primitive Streak and Its Bifurcation into Paraxial and Lateral Mesoderm

(A) Each lineage step labeled with a circled number, corresponding to respective sections in the main text and Figure 7A.

(B) FACS (fluorescence-activated cell sorting) of *MIXL1-GFP* hESC (Davis et al., 2008) after 24 hr in anterior or mid PS induction (left). All cells coexpress *BRACHYURY* and *MIXL1* by scRNA-seq. Each dot depicts a single cell (right).

(C) BMP induces, whereas WNT inhibits, lateral mesoderm from the PS on day 1–2. (i) qPCR of day 1 PS treated with BMP4 or a BMP inhibitor (DM3189) for 24 hr (in the context of A8301 + FGF2 [AF]); (2) qPCR of day 1 PS treated with WNT agonists (CHIR99021 or WNT3A) or WNT inhibitors (300 ng/mL Dkk1 or 1 μM C59) for 24 hr (in the context of A8301 + BMP4 [AB]); error bars = SEM for this and all other qPCR experiments.

(D) BMP inhibits, whereas WNT induces, paraxial mesoderm from the PS on day 1–2. (i) qPCR of day 1 PS treated with BMP4 or a BMP inhibitor (DM3189) for 24 hr (in the context of A8301 + FGF2 [AF]); (ii) qPCR of day 1 PS treated with WNT agonist (3 μM CHIR99021) or WNT inhibitors (2 μM IWR1, 1 μM XAV939 or C59, or 300 ng/mL Dkk1) for 24 hr (in the context of A8301 + DM3189 + FGF2 [ADF]).

(E) Lateral versus paraxial mesoderm bifurcation.

(F) CDX2 and HAND1 staining of day 2 H7-derived paraxial or lateral mesoderm populations or undifferentiated hESCs (scale bar, 100 μm), with Hoechst nuclear staining.

(G) scRNA-seq of day 2 lateral mesoderm or DLL1^{+} sorted paraxial mesoderm. Each dot depicts a single cell. % of marker-positive cells above the dotted TPM (transcripts per million) threshold.

See also Figures S1 and S2.

Yet gaps in our understanding have been revealed by efforts to differentiate human pluripotent stem cells (hPSCs) to various mesoderm cell types in a dish. Human mesoderm has remained obscure because it first forms in gestational weeks 2–4 (O'Rahilly and Müller, 1987), when it is impermissible to access human embryos. There has been some success in generating human mesoderm derivatives from PSCs, including paraxial (Cheung et al., 2012; Mendjan et al., 2014; Umeda et al., 2012) and heart (Ardehali et al., 2013; Burridge et al., 2014; Chong et al., 2014; Lian et al., 2012; Mendjan et al., 2014) cell types. However, because the sequence of lineage branchpoints and the identity of inductive or repressive signals at every developmental step remain incompletely understood, some mesodermal differentiation protocols take weeks to months and generate heterogeneous mixtures of cell types comprising a subset of the desired lineage along with other contaminating lineages. Prior studies indicated that ACTIVIN/NODAL/TGFβ (henceforth referred to as TGFβ), BMP, FGF, and WNT broadly induce mesoderm from PSCs (Cheung et al., 2012; Gertow et al., 2013), the importance of dynamic WNT signaling during cardiac induction (Burridge et al., 2014; Lian et al., 2012; Ueno et al., 2007), and that BMP inhibits paraxial mesoderm formation (Cheung et al., 2012; Umeda et al., 2012). Nonetheless, the challenges faced by current differentiation strategies provide an impetus to better understand the complex process of mesoderm development.

Here, we delineate a roadmap for human mesoderm development and define the sequential steps through which pluripotent cells elaborate a diversity of mesodermal progeny. At many developmental steps, we discovered the minimal combinations of signals sufficient to efficiently induce each human mesodermal fate and showed that it was key to define both *inductive* and *repressive* cues at each step (Table S1). It was critical to define how "unwanted" cell fates were specified in order to logically block their formation and steer stem-cell differentiation down a singular developmental path.

This knowledge guided the efficient differentiation of PSCs into a variety of human mesoderm fates within several days, without recourse to gene modification or serum treatment. The authenticity of the induced cells was confirmed by their ability to engraft in vivo and by single-cell RNA-seq to test for lineage identity and homogeneity. Global RNA-seq and ATAC-seq (assay for transposase-accessible chromatin with high throughput sequencing) analyses revealed the stepwise changes in gene expression and sequential opening and closing of chromatin elements at each developmental transition. Collectively, we chart the developmental landscape of human mesoderm formation and uncover the sequential signaling, transcriptional, and chromatin changes at each lineage step. We directly demonstrate the utility of this reference map in guiding stem-cell differentiation, producing transplantable cells for eventual use in regenerative medicine, improving our understanding of human development, and uncovering the putative origins of certain human congenital malformations.

RESULTS

Induction of Anterior and Mid Primitive Streak

Primitive streak (PS) formation from pluripotent cells is the first step in mesoderm development (Figure 1A, step 1). We generated a >98% pure MIXL1-GFP$^+$ human PS population within 24 hr of PSC differentiation (Figures 1Bi, S1A, and S1B) by activating TGFβ, WNT, and FGF and inhibiting PI3K signaling, in the presence or absence of exogenous BMP (Figures S1C–S1E) (Gertow et al., 2013; Loh et al., 2014; Schier and Talbot, 2005; Tam and Loebel, 2007). Attesting to the uniformity of differentiation, single-cell RNA-seq (scRNA-seq) of bulk PS populations revealed that 100% of analyzed cells coexpressed PS TFs *BRACHYURY* and *MIXL1* (Figure 1Bii).

In the vertebrate embryo, different anterior-posterior regions of the PS produce distinct mesoderm derivatives (Lawson et al., 1991; Rosenquist, 1970; Tam and Beddington, 1987). Likewise, hPSC-derived anterior PS induced in the presence of an anteriorizing TGFβ signal was competent to form paraxial mesoderm (Figure S1E). By contrast, mid PS induced in the presence of both anteriorizing TGFβ and posteriorizing BMP harbored maximal potential to form lateral mesoderm/cardiac progenitors (Figures S1E–S1I). Thus, as in model organisms, human PS is not a single lineage but comprises several subtypes each already partially committed to one type of downstream mesoderm (Figures S1J–SL). Together, these >98% pure human PS populations provided a starting point to understand the subsequent divergence of distinct mesoderm subtypes.

Bifurcation of Paraxial versus Lateral Mesoderm from Primitive Streak by Competing WNT and BMP Signals

In vivo, the PS forms definitive endoderm, paraxial mesoderm, and lateral mesoderm, but how these lineages are segregated is not well understood (Figure 1A, step 2). After PS induction (day 0–1), TGFβ specified endoderm (Figures S2A and S2B) (Loh et al., 2014) while TGFβ inhibition blocked endoderm formation and instead induced mesoderm (Figure S2B–S2E). Since TGFβ inhibition and FGF/ERK activation (Figures S2B and S2F) for 24 hr (day 1–2) created a permissive context for both paraxial and lateral mesoderm formation, we sought how these mutually exclusive mesoderm subtypes become distinguished.

Countervailing BMP and WNT signals respectively induced human lateral versus paraxial mesoderm and each repressed the formation of the mutually exclusive lineage on day 1–2 of PSC differentiation, driving the bifurcation of these two mesoderm subtypes from the PS (summarized in Figure 1E). Exogenous BMP induced lateral mesoderm and repressed paraxial mesoderm (Figures 1Ci, 1Di, S2E, and S2H). By contrast, blocking BMP signaling abrogated lateral mesoderm and instead expanded paraxial mesoderm (Figures 1Ci, 1Di, and S2H). This reveals a key function of BMP in patterning human mesoderm akin to its activity in chick (Tonegawa et al., 1997).

Conversely, WNT promoted human paraxial mesoderm and repressed lateral mesoderm. WNT activation (by GSK3 inhibition) induced paraxial markers while suppressing lateral/cardiac markers (Figures 1Cii and 1Dii). By contrast, WNT inhibition elicited lateral mesoderm while blocking the paraxial fate (Figures 1Cii and 1Dii). Therefore, WNT controls the allocation of human paraxial versus lateral mesoderm, logically linking the requirement for WNT in mouse paraxial mesoderm formation (Aulehla et al., 2008) to the ability of WNT to repress early cardiac mesoderm in *Xenopus* (Schneider and Mercola, 2001).

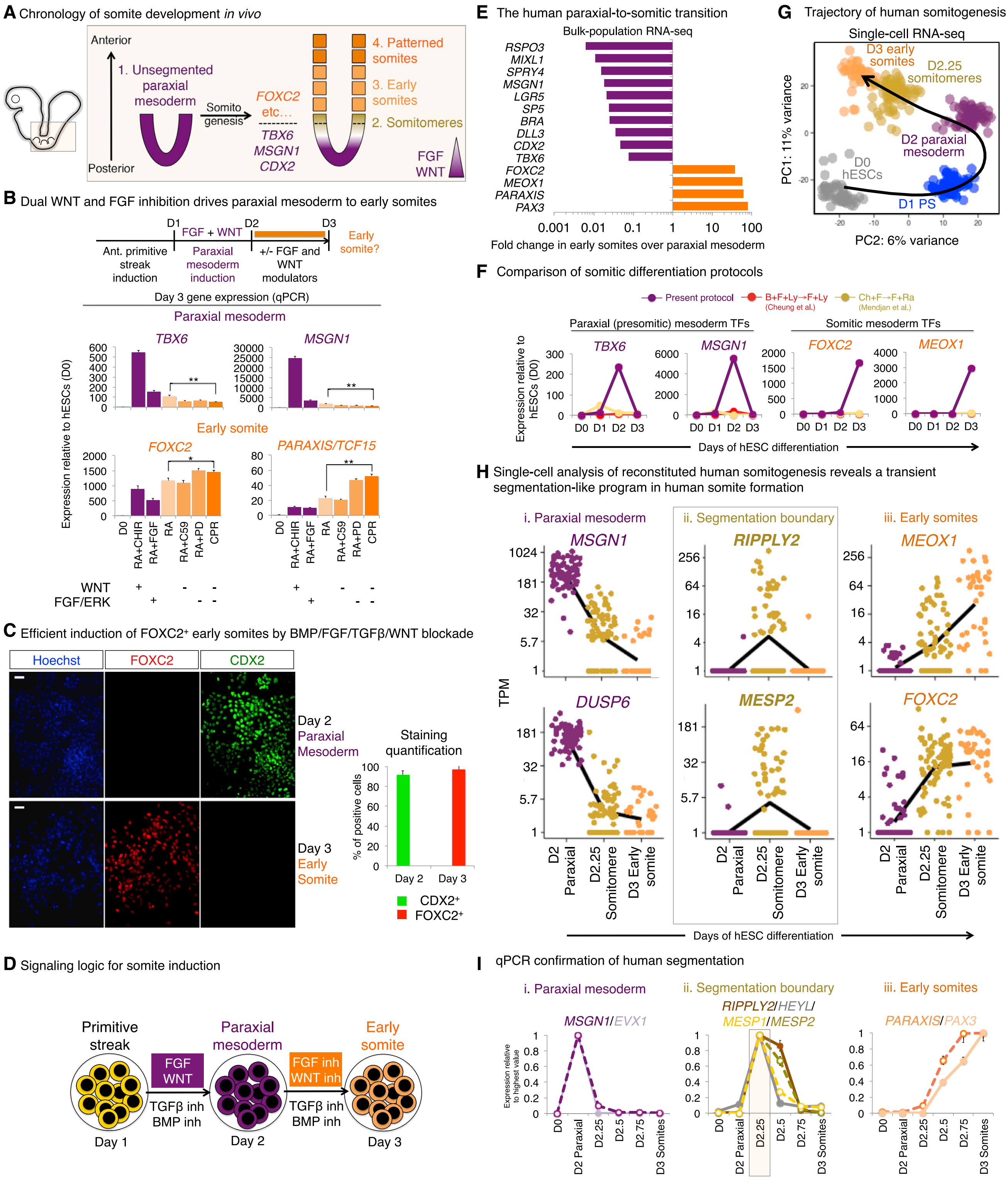

Figure 2. Human Paraxial Mesoderm Differentiation into Early Somites Passes through an Ephemeral Somitomere-Like State

(A) Paraxial mesoderm segmentation into somites in vivo.

(B) To reveal how WNT and FGF/ERK control paraxial mesoderm progression to early somites, day 2 H7-derived paraxial mesoderm was treated with RA (2 μM) for 24 hr, in combination with a WNT agonist (CHIR, 3 μM), a WNT inhibitor (C59, 1 μM), FGF2 (20 ng/mL), an ERK inhibitor (PD0325901, 500 nM), or combined

(legend continued on next page)

In summary, on day 1–2 of hPSC differentiation, BMP inhibition and WNT activation induced paraxial mesoderm whereas, conversely, BMP activation and WNT inhibition specified lateral mesoderm from the PS within the permissive context of TGFβ inhibition/FGF activation (Figure 1E). These two mutually exclusive signaling conditions produced either $CDX2^{+}HAND1^{-}$ paraxial mesoderm or $CDX2^{lo/-}HAND1^{+}$ lateral mesoderm by day 2 of differentiation (Figure 1F and S2I). Tracking the bifurcation of paraxial versus lateral mesoderm fates by scRNA-seq confirmed the mutually exclusive marker expression in the two diverging populations at the level of single cells (Figure 1G). scRNA-seq showed that both human paraxial and lateral mesoderm populations were essentially uniform: 98.5% of paraxial mesoderm cells expressed *DLL3* and *MSGN1* whereas 98.1%–100% of lateral mesoderm cells expressed *HAND1* and *FOXF1* (Figure 1G).

Maturation of Paraxial Mesoderm into Early Somites by Combined BMP, ERK, TGFβ, and WNT Inhibition

Having generated 91.2% ± 0.1% pure $TBX6^{+}CDX2^{+}$ human paraxial mesoderm by day 2 of PSC differentiation (Figure S3A), we next sought to drive these cells into early somite progenitors (Figure 1A, step 3, and Figure 2A). During embryogenesis, the U-shaped sheet of paraxial (presomitic) mesoderm is progressively segmented at its anterior edge to generate spherical early somites (Figure 2a), due to lower anterior levels of FGF and WNT at the "wavefront" (Aulehla et al., 2008; Dubrulle et al., 2001).

Whereas paraxial mesoderm was specified on day 1–2 by FGF and WNT, the inhibition of FGF/ERK and WNT signaling on day 2–3 strongly downregulated paraxial mesoderm genes (e.g., *TBX6*, *MSGN1*) and upregulated early somite markers (e.g., *FOXC2*, *PARAXIS*, *MEOX1*; Figures 2B and S3B). Early somite markers were further upregulated when TGFβ and BMP were inhibited (Figures S3C and S3D); therefore, we employed quadruple inhibition of these four pathways to drive near-complete conversion of day 2 $CDX2^{+}$ paraxial mesoderm into 96.8% ± 5.7% pure $FOXC2^{+}$ early somite precursors by day 3 (Figures 2C–2E). Together, this identified the minimal signaling cues sufficient to efficiently generate human PS, paraxial mesoderm, and subsequently, early somite progenitors from PSCs within 72 hr of differentiation (Figure 2D) more robustly and rapidly than was previously possible (Cheung et al., 2012; Mendjan et al., 2014) (Figure 2F).

scRNA-seq reaffirmed the homogeneity of these various in-vitro-derived populations at different steps of human somitogenesis. 98.5%–100% of human paraxial mesoderm cells (sorted for DLL1 expression; Supplemental Experimental Procedures) expressed paraxial markers *MSGN1* and *DUSP6* at day 2, yet these markers were sharply suppressed within 6 hr of FGF/WNT inhibition (by day 2.25) during differentiation toward somites (Figure 2Hi). Conversely, somite TFs *MEOX1* and *FOXC2* became expressed in the majority (83.3%–88.9%) of human early somite cells by day 3 (Figure 2Hiii). Hence, this indicated that human somitogenesis was efficiently reconstituted in culture. We exploited this system to uncover developmental features of this process.

Single-Cell RNA-Seq Identifies a Conserved, Segmentation-Like Process in Human Somitogenesis and Infers Transcriptional Cofactor *HOPX* as a Marker

In vertebrate model organisms, somitogenesis entails transient expression of segmentation genes (for 30–120 min in fish and mice, respectively) at the wavefront, inducing paraxial mesoderm to segment into somitomeres (prospective somites) and then into somites (Pourquié, 2011). Human somitogenesis begins at week 3 of gestation (O'Rahilly and Müller, 1987), but whether it also entails a transient segmentation molecular program has remained unclear because it is ethically impermissible to access or analyze week 3–5 human embryos.

Indeed, paraxial mesoderm cells in our reconstituted human somitogenesis system passed through an intermediate somitomere stage before differentiating into somites. scRNA-seq of human embryonic stem cells (hESCs) (day 0) differentiating into anterior PS (day 1), paraxial mesoderm (day 2), through an intermediate step (day 2.25), and into early somites (day 3) showed that single cells uniformly moved through a single inferred differentiation trajectory without overtly diverging branch points (Figure 2G), implying the homogeneity of cells at each lineage step. Principal component analysis (PCA) positioned the day 2.25 population between day 2 paraxial mesoderm and day 3 early somites (Figure 2G), suggesting that it is a true intermediate between presomitic/paraxial and somitic states.

Indeed, a subset of day 2.25 cells transiently expressed *MESP2*, *RIPPLY2*, and *HEYL* (Figures 2Hii and S4E), whose homologs are somitomere segmentation markers in model organisms (reviewed by Pourquié, 2011). qPCR confirmed a brief pulse of *MESP2*, *RIPPLY2*, and *HEYL* expression for several hours (~day 2.25–day 2.5) in the interval between paraxial and somitic marker expression (Figure 2I). Hence, there exists a transient somitomere-like transition point in human somitogenesis, arguing that human development entails an evolutionarily conserved segmentation process.

To discover additional somite segmentation markers, single-cell transcriptomes were aligned in "pseudotime" to infer the

WNT/ERK inhibition (C59+PD0325901+RA [CPR]) and qPCR was conducted (*$p < 0.05$, **$p < 0.01$), showing WNT/ERK blockade enhances early somite induction (it was later found that exogenous RA was dispensable for early somite formation; Figure S3E).

(C) CDX2 and FOXC2 staining of BJC1-derived paraxial mesoderm (day 2) and early somite (day 3) populations (left) and quantification (right).

(D) FGF and WNT activation, followed by inhibition, induces human paraxial mesoderm and then early somites.

(E) Differentially expressed genes in day 2 paraxial mesoderm versus day 3 early somites (bulk-population RNA-seq).

(F) qPCR time-course comparison of H7 hESCs differentiated into somites using previous protocols (Cheung et al., 2012; Mendjan et al., 2014) or the current method.

(G) PCA of human somitogenesis scRNA-seq. Colors designate cell populations harvested at different time points. Each dot is a single cell.

(H) scRNA-seq of day 2, day 2.25, and day 3 hESC-derived populations. Dots depict single cells. Line indicates mean gene expression in all cells at each time point.

(I) Time.-course qPCR of H7-derived cells.

See also Figure S3.

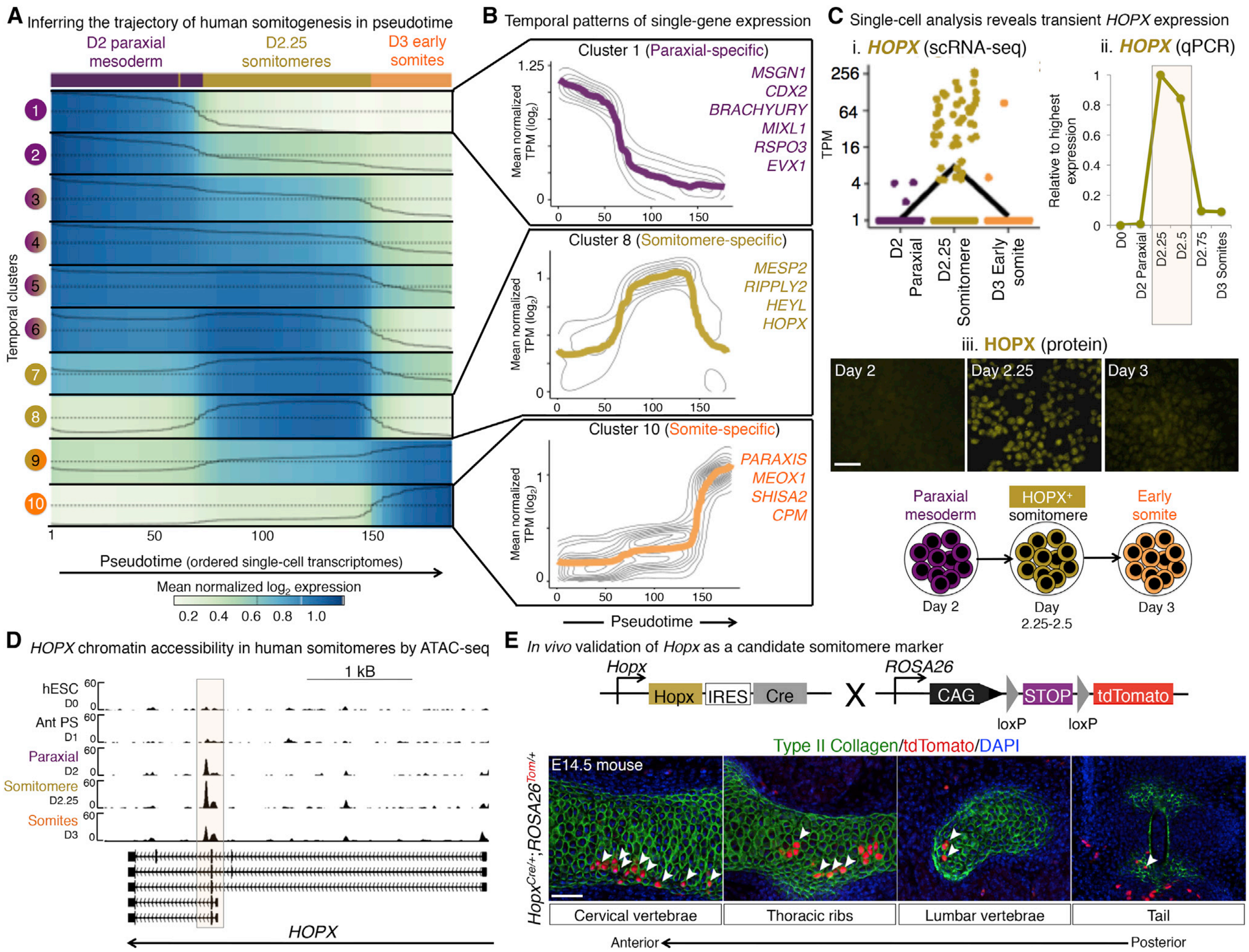

Figure 3. Single-Cell Analysis Captures a Transient *HOPX*$^+$ Human Somitomere Progenitor State

(A) Heatmap of normalized scRNA-seq gene expression across the inferred trajectory of human somitogenesis. Each column reflects a single cell, with scRNA-seq paraxial mesoderm, somitomere, and early somite transcriptomes (colored blocks) ordered in pseudotime along the y axis (Supplemental Experimental Procedures). Genes were clustered into ten clusters (rows) by virtue of their expression kinetics across this pseudotime time course. Line indicates smoothed mean expression of all genes in the cluster across pseudotime.

(B) Mean expression (bold line) of all genes in each temporal cluster across pseudotime (with contours representing density of individual gene expression), with representative genes in each cluster noted.

(C) Transient *HOPX* expression during H7 hESC differentiation toward somites, shown by scRNA-seq (1), qPCR (2), and immunostaining (3); scale bar, 50 μm.

(D) ATAC-seq shows the *HOPX* locus is accessible in day 2.25 hESC-derived somitomeres (signal track, $-\log_{10}$ p values).

(E) Fate mapping progeny of *Hopx*$^+$ cells in E14.5 *Hopx-IRES-Cre*;*Ai9* embryos reveals contribution to the spine and ribs (labeled by type II collagen); scale bar, 50 μm.

order in which they arose along an inferred developmental path (Supplemental Experimental Procedures; Figure 3A). This led to 44 candidate genes with transient somitomere-specific expression (Figure 3B), including *HOPX*, a homeodomain-containing transcriptional cofactor. By scRNA-seq, *HOPX* was specifically expressed in a subset of day 2.25 somitomere cells but neither day 2 paraxial mesoderm nor day 3 early somites (Figure 3Ci). qPCR (Figure 3Cii) and immunostaining (Figure 3Ciii) corroborated the transient expression of *HOPX* mRNA and protein during in vitro somitogenesis, paralleling the transitory expression of known somitomere markers (Figure 2I). Additionally, open chromatin analysis by ATAC-seq (Buenrostro et al., 2013) revealed that the *HOPX* locus was accessible in day 2.25 somitomere populations (Figure 3D). Thus, *HOPX* is briefly expressed in somitomeres and marks human somite segmentation.

Consistent with the notion that *Hopx* also marks somitomeres in mouse embryos, genetic lineage tracing showed that *Hopx*$^+$ cells contributed to the ribcage, cervical, and lumbar vertebrae and tail (Figure 3E). Sparse labeling reflects the noted inefficiency of the *Hopx-IRES-Cre* driver allele (Jain et al., 2015). Collectively, this uncovers a transient molecular signature of somite segmentation conserved between human and other vertebrates and suggests *HOPX* as a possible component of this signature.

Bifurcation of Human Early Somite Progenitors into Sclerotome and Dermomyotome by HH and WNT Signals

After human somites are formed, how do they diverge into distinct derivatives (Figure 1A, step 4)? In vivo, early somites are patterned along their dorsal-ventral axis to generate sclerotome (ventral somite; precursor to bone, cartilage, and smooth muscle) and dermomyotome (dorsal somite; precursor to skeletal muscle, brown fat, and dorsal dermis; Figure 4A).

Starting from day 3 human early somite progenitors, HEDGEHOG (HH) induced sclerotome while WNT specified dermomyotome and each cross-antagonized the effect of the other on days 3–5 of differentiation (Figures 4B and S4A and as summarized in Figure 4H). By activating one signal while inhibiting the other, it was possible to specify one somitic derivative while blocking formation of the alternate fate.

HH activation together with WNT inhibition blocked dermomyotome formation, inducing a fairly uniform SOX9^{+}TWIST1^{+} sclerotome population that expressed multiple sclerotome markers (*PAX1*, *PAX9*, *NKX3.2/BAPX1*, *FOXC2*, *SOX9*, and *TWIST1*) (Figures 4B, 4C, S4A, and S4B). Conversely, WNT activation together with HH blockade blocked sclerotome formation and instead exclusively specified dermomyotome (Figures 3B and S4A). If HH and WNT were simultaneously activated, neither sclerotome or dermomyotome was elicited (Figure S4A), indicating the importance of their mutually exclusive activation. Collectively, this shows that HH and WNT pattern human somites, analogous to their role in mice (Fan et al., 1995, 1997; Fan and Tessier-Lavigne, 1994), and demonstrates how their manipulation can separately generate human sclerotome or dermomyotome. Indeed, scRNA-seq showed that hPSC-derived sclerotome and dermomyotome populations formed in these two mutually exclusive signaling conditions constituted largely distinct clusters (Figure 4D) that diverged downstream of a common early somite progenitor state.

Human dermomyotome induction from early somites relied on WNT activation and initial HH inhibition (to block sclerotome induction) in addition to BMP (to induce *PAX7*; Figure S4C) and later-stage HH activation (to induce *EN1*; Figures S4D–S4F). In vivo, dermomyotome gives rise to skeletal muscle, and accordingly, human dermomyotome was also capable of generating MYH3^{+} skeletal muscle cells in vitro (Figure S4E).

Human Sclerotome Forms an Ectopic Bone In Vivo

Strikingly, human sclerotome (ventral somite) progenitors could generate an ectopic human bone in vivo (Figure 1a, step 5), reflecting the skeleton-forming fate of embryonic sclerotome (Christ and Scaal, 2008). Upon subcutaneous injection into immunodeficient mice, hESC-derived sclerotome formed ectopic bone-like structures containing both bone and cartilage (Figure 4Ei). Other tissues (i.e., epithelia) were not present (Figure S5A), indicating the developmental lineage restriction of sclerotome. Labeling sclerotome with constitutive *GFP* (Figure S5B), *BCL2-2A-GFP* (Figure S5C), or *Luciferase* (Figure S5D) reporters prior to transplantation confirmed that the subcutaneous GFP^{+} or luciferase^{+} bones (Figure 4E) were not derived from resident mouse cells.

Sclerotome-derived ectopic human bones self-organized themselves even though they were implanted without a patterned matrix. Specifically, they harbored proliferative chondrocytes that progressed to hypertrophic chondrocytes and finally underwent ossification in a spatially choreographed fashion within column-shaped streams of cells (Figure 4F). They also recruited host blood vessels (Figure S5E), recapitulating a developmental endochondral ossification program. In summary, hESC-derived sclerotome harbored bone/cartilage progenitor activity in vivo and formed bones in a process mimicking natural bone development.

Human sclerotome could also generate cartilage and fibroblasts in vitro (Figure 1A, step 6). Exposure to BMP (Murtaugh et al., 1999) upregulated cartilage structural genes, yielding a fairly uniform COL2A1^{+} population that later secreted glycosaminoglycans (Figures 4Gi and S5E). By contrast, PDGF and TGFβ treatment (Cheung et al., 2012) for 3 days yielded a >90% pure SMAα^{hi} fibroblast-like population (Figures 4Gii and S5E). Having defined the signaling logic for PS differentiation into human paraxial mesoderm and somitic derivatives, next we focused on the parallel lineage branch: PS differentiation into lateral mesoderm and cardiac fates (Figure 1A, step 7).

Lateral Mesoderm Patterning into Heart versus Limbs by Competing FGF and WNT Signals

We sought to define how human lateral mesoderm is diversified into multiple derivatives, including *NKX2.5*$^{+}$ heart-forming anterior-lateral mesoderm and limb-forming *PRRX1*$^{+}$ posterior-lateral mesoderm (Figure 1A, step 7, and Figure 5A) (Tanaka, 2013).

Bifurcation of day 2 lateral mesoderm into day 3 anterior (cardiac) versus posterior (limb bud) fates was respectively induced by FGF and WNT signals (summarized in Figure 5A). WNT posteriorized lateral mesoderm, inducing limb markers *PRRX1* and *HOXB5* while suppressing heart field markers *NKX2.5* and *TBX20* on day 2–3 of hESC differentiation (Figures 5B and S6A). Reciprocally, WNT inhibition suppressed posterior-lateral mesoderm and instead induced cardiac mesoderm (Figures 5B and S6A). Thus, our findings corroborate the clear requirement for WNT blockade for cardiac specification in vivo (Schneider and Mercola, 2001) and in vitro (Burridge et al., 2014; Lian et al., 2012; Mendjan et al., 2014) and further indicate that WNT inhibitors induce heart precursors by restraining limb formation (Figure S6D).

Conversely, FGF anteriorized human lateral mesoderm. Exogenous FGF enhanced NKX2.5-GFP^{+} cardiac mesoderm induction on day 2–3 of hESC differentiation, whereas FGF inhibition abolished cardiac induction (Figure 5C). This demonstrates a conserved role for FGF in cardiac mesoderm specification from humans to zebrafish (Reifers et al., 2000).

Hence, activating pro-cardiac FGF signaling and inhibiting pro-limb WNT signaling (in the permissive context of BMP activation and TGFβ inhibition) efficiently directed day 2 lateral mesoderm toward cardiac mesoderm, as tracked using an *NKX2.5-GFP* knockin hESC line (Elliott et al., 2011). We respectively obtained >80% and >90% NKX2.5-GFP^{+} cardiac mesoderm by days 3 and 4 of hPSC differentiation (Figure 5D), which was more rapid and robust than a WNT-modulator-only cardiac induction protocol (Figure 5E) (Burridge et al., 2014).

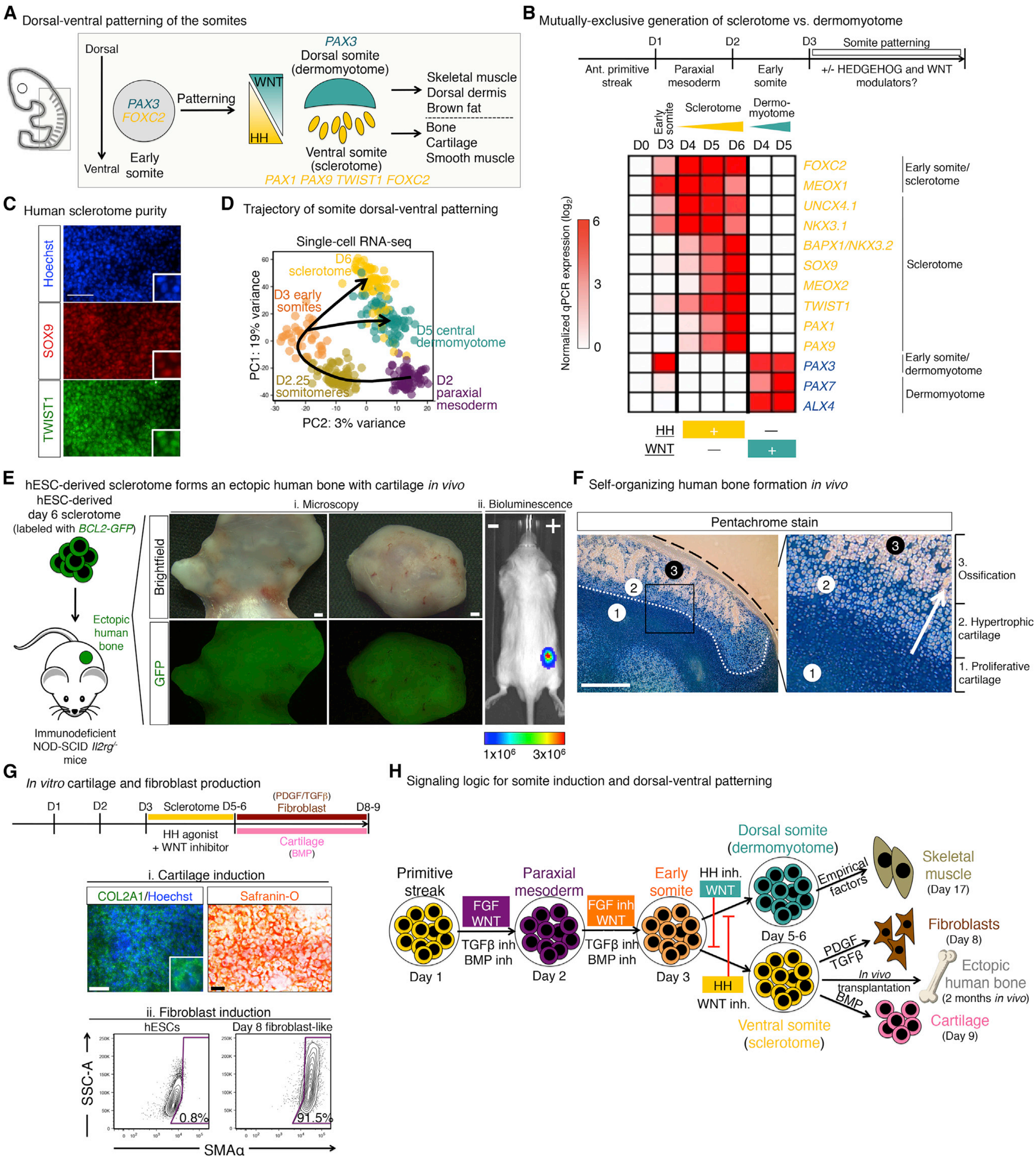

Figure 4. Dorsal-Ventral Patterning of Somite Precursors into Sclerotome and Dermomyotome and Downstream Progeny

(A) Somite patterning in vivo.

(B) qPCR heatmap of hESCs (day 0), early somite progenitors (day 3), or those differentiated into sclerotome (day 4, day 5, or day 6, using 21K+C59) or dermomyotome (day 4 or day 5, using BMP4+CHIR+Vismodegib).

(C) SOX9 and TWIST1 staining of day 6 H7-derived sclerotome; scale bar, 100 μm.

(D) PCA of scRNA-seq from indicated populations; each dot depicts a single cell.

(E) *EF1A-BCL2-2A-GFP* expressing H9-derived sclerotome was subcutaneously injected into NSG mice; 2 months later, ectopic GFP^{+} human bones formed (left); bioluminescent imaging of mice 1 month post-transplantation by *UBC-Luciferase-2A-tdTomato* H9-derived sclerotome.

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Directing Cardiac Mesoderm into Human Cardiomyocytes that Can Engraft Human Fetal Heart In Vivo

Having rapidly generated a >90% pure NKX2.5^{+} cardiac mesoderm population by day 4 of hESC differentiation (Figures 5D and 5E), we next sought to drive these progenitors toward cardiomyocytes (Figure 1A, step 8). Single cardiac mesoderm progenitors can form both myocardium (cardiomyocytes) and endocardium in mice (Devine et al., 2014).

BMP activation together with low FGF levels preferentially induced cardiomyocytes from cardiac mesoderm (Figures S6E and S6F) at the expense of (pro)epicardium or endocardium (Figure S6E). WNT activation seemed to sustain undifferentiated *ISL1*$^{+}$ cardiac progenitors and inhibited maturation into cardiomyocytes (Figure S6E). Therefore, WNT blockade in conjunction with BMP activation enhanced cardiomyocyte differentiation (Figure S6H), which was enhanced by vitamin C (Figure S6I) (Burridge et al., 2014). Treating day 4 cardiac mesoderm with these factors yielded a 72.2% ± 5.6% and a 77.8% ± 1.6% pure TROPONIN^{+} cardiomyocyte population by days 6 and 8 of hPSC differentiation, respectively (Figure 5F), which spontaneously contracted and expressed cardiomyocyte structural genes (Figure S6J).

The authenticity of hESC-derived cardiac lineages was confirmed by their ability to engraft human fetal heart tissue. hESC-derived cardiomyocytes can engraft model organisms (e.g., guinea pigs and non-human primates; Chong et al., 2014) but therapies will require evidence that such cells can engraft *human* heart tissue. To this end, we employed an experimental system whereby ventricular fragments from week 15–17 human fetal heart (Ardehali et al., 2013) were subcutaneously implanted into the mouse ear (Figure 5G). These human heart fragments were revascularized and continued beating for months in vivo, as shown by QRS electrocardiogram signals (Figure 5G). Upon transplantation of luciferase^{+}/GFP^{+} hESC-derived heart populations, both day 3 cardiac mesoderm and day 8 cardiomyocytes engrafted human ventricular fragments for at least 10 weeks, as indicated by bioluminescence imaging (n = 10 successfully engrafted human heart fragments obtained from 2 fetal donors; Figure 5Hi). Within human fetal heart tissue, GFP^{+} hESC-derived cardiomyocytes expressed cytoskeletal protein TROPONIN/TNNT2 and membranous gap junction protein CONNEXIN 43 (Figure 5Hii). Together, these data show that our differentiation strategy can lead to the rapid generation of a ~75% pure cardiomyocyte population capable of engrafting human fetal heart tissue.

Identifying Cell-Surface Markers to Allow Purification of Diverging Human Mesoderm Subtypes

After generating this hierarchy of mesoderm lineages, we also defined lineage-specific cell-surface markers to track different mesoderm lineages and enable purification of each mesoderm lineage for assessment of biological function and fate and for potential therapeutic purposes in the future. Screening 332 cell-surface markers across hESCs and 7 mesodermal lineages (Figure 6A; Table S2) revealed that certain previously described mesoderm markers were broadly expressed in both paraxial and cardiac mesoderm (Figures S7A and S7B). Therefore, we sought lineage-specific markers.

Surface markers DLL1 and GARP respectively marked paraxial mesoderm and cardiac mesoderm in a mutually exclusive fashion. NOTCH ligand DLL1 was specifically expressed in day 2 paraxial mesoderm, whereas conversely, GARP marked day 3 NKX2.5-GFP^{+} cardiac mesoderm. Neither marker was expressed by undifferentiated hESCs (Figures 6B and S7C), thus tracking a clear bifurcation of PS into paraxial or cardiac mesoderm fates.

DLL1 and GARP were mesodermal markers conserved from human to zebrafish. GARP/LRRC32, a transmembrane protein that tethers TGFβ ligands to the cell surface (Tran et al., 2009), was likewise expressed in the heart tube of zebrafish embryos (Figure 6C). Conversely, *deltaC* (a homolog to human *DLL1*) was likewise expressed in zebrafish paraxial mesoderm (Figure S7D), as reported previously (Smithers et al., 2000).

These surface markers enabled the purification of desired mesoderm lineages from admixed cultures, providing tools to refine stem-cell differentiation. DLL1 was expressed by 91.6% ± 5.4% of cells after 2 days of paraxial mesoderm induction. Paraxial mesoderm-specific TFs (*TBX6*, *MSGN1*) were exclusively expressed by the DLL1^{+} fraction (Figure 6D). Sorted DLL1^{+} human paraxial mesoderm cells were essentially uniform as shown by scRNA-seq: 97.0% of cells coexpressed archetypic paraxial markers *MSGN1* and *DLL3* (Figure 6E). *TBX6* mRNA expression in all but a few cells (Figure 6E) may reflect technical dropout in scRNA-seq (Marinov et al., 2014). Hence, sorting for DLL1^{+}GARP^{-} cells purifies human paraxial mesoderm attained from either differentiating ESCs or iPSCs (induced pluripotent stem cells; Figure S7E), providing a method to isolate pure human paraxial mesoderm and interrogate its characteristics.

Downstream of paraxial mesoderm during the bifurcation of sclerotome versus dermomyotome fates, surface marker PDGFRα enabled the purification of sclerotome. PDGFRα was expressed by 85.2% ± 8.4% of cells in day 5–6 sclerotome cultures, and only the PDGFRα^{+} fraction expressed sclerotome markers (*FOXC2*, *PAX1*, and *PAX9*), consistent with *pdgfrα* expression in the sclerotome, but not dermomyotome of zebrafish embryos (Figure 6F) (Liu et al., 2002). In vitro, PDGFRα was indeed expressed at higher levels in hPSC-derived sclerotome relative to dermomyotome (Figure S7E), thereby helping to distinguish ventral from dorsal somite fates.

scRNA-seq of PDGFRα^{+} human sclerotome revealed 86.2% of cells coexpressed chondroprogenitor markers *SOX9* and

(F) Russell-Movat's Pentachrome staining of 2-month-old sclerotome grafts revealed zones of chondrogenesis and ossification, with cartilage stained blue; black line denotes the edge of the graft; white line denotes boundary of the ossifying region; scale bar, 1 mm (bottom).

(G) COL2A1 (top left) and Safranin-O staining (top right) of day 6+2 or day 6+6 hESC-derived cartilage, respectively; scale bars, 0.1 mm (left) and 1 mm (right); SMAα intracellular FACS of hESCs or day 8 fibroblast-like cells (bottom).

(H) Somite patterning into dermomyotome or sclerotome and downstream differentiation.

See also Figures S4 and S5.

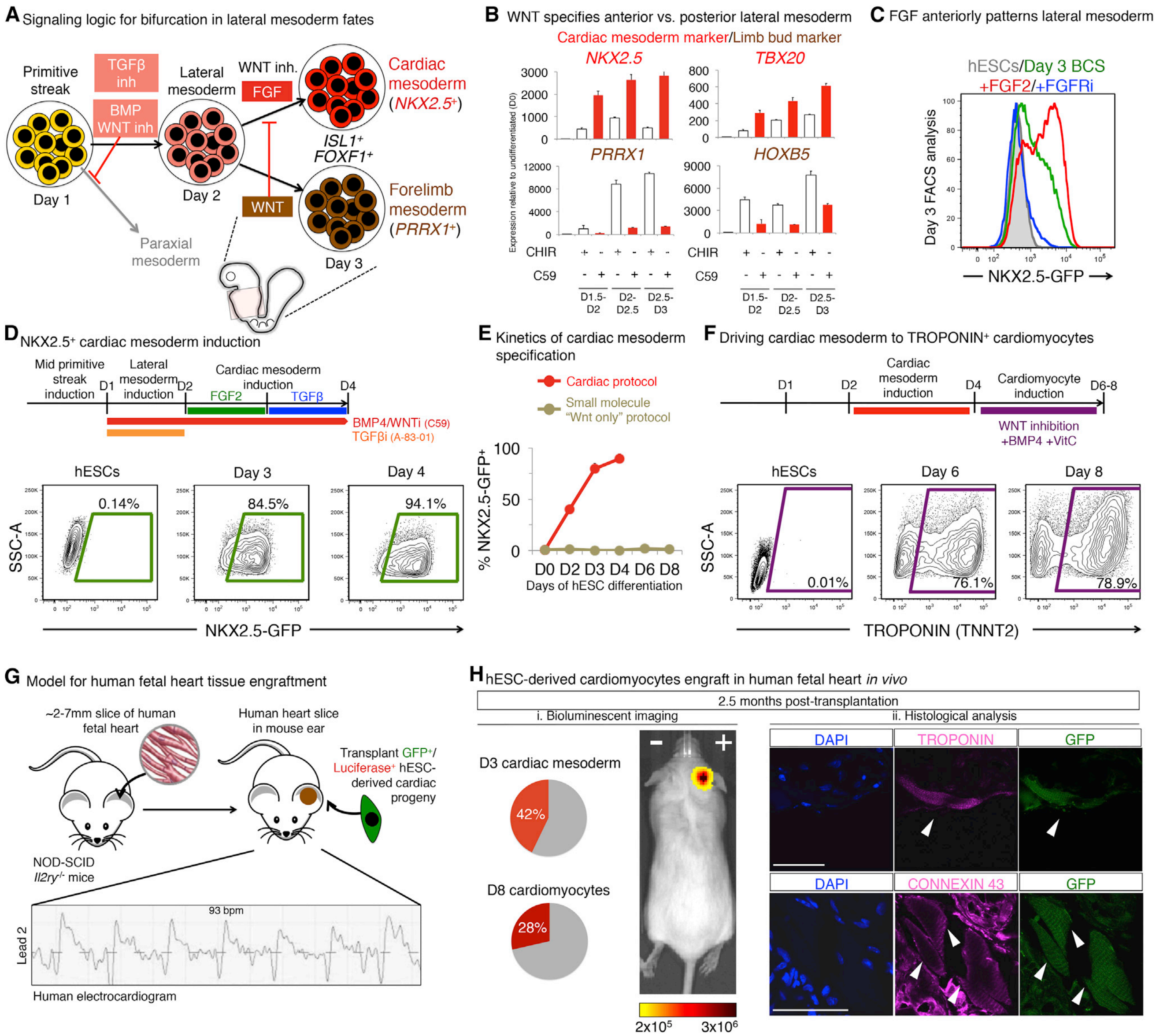


Figure 5. Lateral Mesoderm Patterning into Cardiac versus Limb Mesoderm Fates

(A) Cardiac versus forelimb bifurcation.

(B) To assess the role of WNT in lateral mesoderm patterning, day 1 PS was differentiated to lateral mesoderm (30 ng/mL BMP4 + 1 μM C59 + 2 μM SB505124 [BS]) for varying lengths of time (until day 2, day 2.5, or day 3) and, for the last 12 hr, was treated with C59 or 3 μM CHIR (in addition to BS), and qPCR was conducted.

(C) To assess the role of FGF in lateral mesoderm patterning, day 2 *NKX2.5-GFP* lateral mesoderm was treated with BMP4 + C59 + SB505124 (BCS) with or without FGF2 (20 ng/mL) or FGFR inhibitor PD173074 (100 nM) for 24 hr, and FACS was conducted on day 3.

(D) Time-point FACS of *NKX2.5-GFP* hESC (Elliott et al., 2011) differentiation using cardiac mesoderm protocol.

(E) Comparison of NKX2.5-GFP$^+$ cell percentages (determined by FACS) on days of differentiation, using the current protocol or a previous method (Burridge et al., 2014).

(F) Intracellular TNNT2 FACS of H7-derived cardiomyocytes (*bottom*).

(G) Electrocardiogram of human fetal heart implanted in the mouse ear, >1 month post-implantation.

(H) 2.5 months post-transplant of *EF1A-BCL2-2A-GFP*;*UBC-tdTomato-Luciferase* H9 hESC-derived cardiac lineages into human fetal heart grafts, luciferase$^+$ donor cells were detected (i); engrafted hESC-derived cardiomyocytes were TROPONIN$^+$ and CONNEXIN 43$^+$; scale bar, 40 μm (ii).

See also Figure S6.

COL2A1 (Figure 6G), reaffirming the skeletal stem-cell-like nature of these cells and consistent with their phenotypic ability to form ectopic bone grafts (see above). However, *PAX9* and *FOXC2* expression was more heterogeneous (Figure 6G), which may reflect distinct anterior-posterior and medial-lateral sclerotome subdomains (Christ and Scaal, 2008).

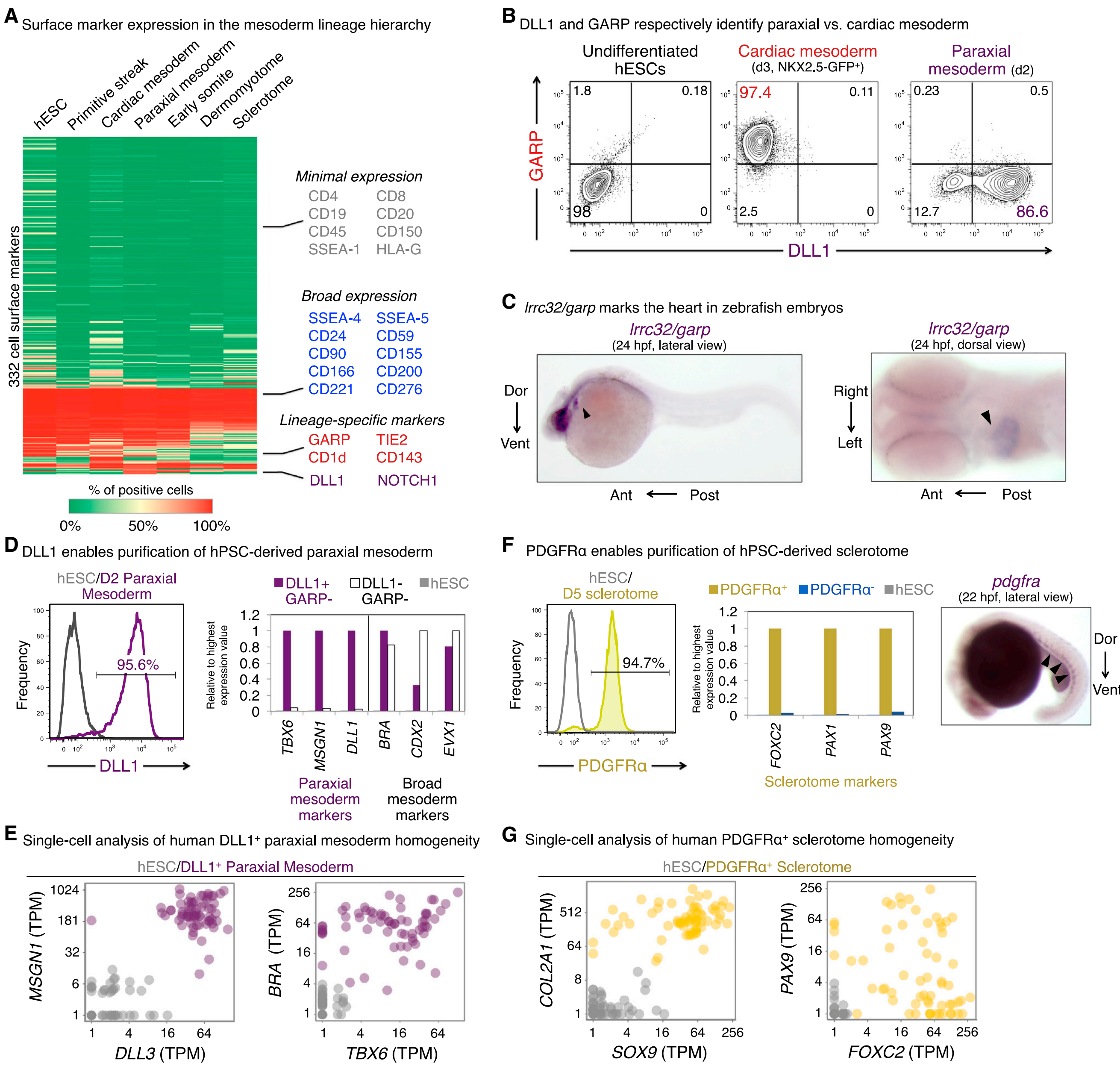


Figure 6. High-Throughput Screen for Lineage-Specific Mesoderm Surface Markers

(A) Clustered heatmap of surface marker expression in hESCs and 6 mesoderm derivatives. Each row represents an individual surface marker and color denotes the percentage of cells positive for a given marker. For PS and cardiac mesoderm, marker expression was analyzed after pre-gating on MIXL1-GFP^{+} and NKX2.5-GFP^{+} fractions, respectively.

(B) GARP and DLL1 FACS in hESCs, day 2 paraxial mesoderm cultures or day 3 NKX2.5-GFP^{+} pre-gated cardiac mesoderm.

(C) *lrrc32* in situ hybridization in 24 hr post-fertilization zebrafish embryo (arrows denote heart).

(D) DLL1 FACS of day 2 paraxial mesoderm culture (left); qPCR of sorted populations (right).

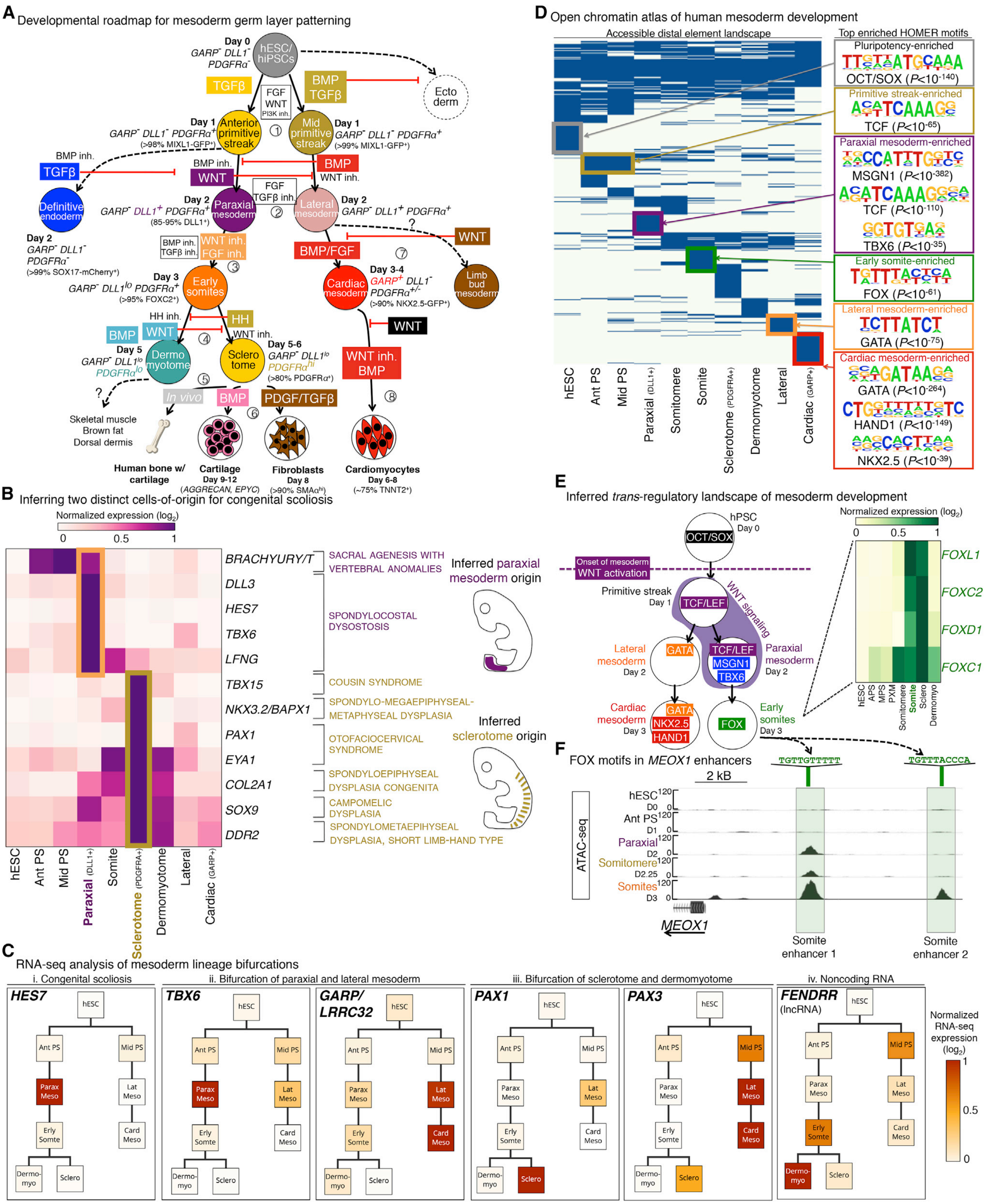
(E) scRNA-seq of sorted DLL1^{+} human paraxial mesoderm; each dot is a single cell.

(F) PDGFRα FACS of day 5 sclerotome population (left); qPCR of sorted PDGFRα^{+} and PDGFRα^{-} populations (center); in situ hybridization for *pdgfra* expression (right) in 22 hpf zebrafish embryo (arrowheads denote ventral staining in sclerotome).

(G) scRNA-seq of sorted PDGFRα^{+} human sclerotome; each dot is a single cell.

See also Figure S7 and Table S2.

In summary, these cell-surface markers define a roadmap for mesoderm development (Figure 7A) by identifying mutually exclusive types of mesoderm progenitors, thus enabling one to track the products of key developmental branchpoints. These markers are likewise expressed by the same cell types in zebrafish embryos (Figures 6 and S7). These markers allowed us to



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confirm that hPSC differentiation to various mesoderm lineages was efficient (Figure 6) and enabled further purification of desired lineages for transcriptional and chromatin analysis.

Global Transcriptional Profiling of the Mesoderm Lineage Hierarchy Confirms Lineage Demarcations and Suggests a Cell-of-Origin for Congenital Malformations

To chart a molecular roadmap for mesoderm development, we used bulk-population RNA-seq to capture global transcriptional dynamics during the commitment of ESCs to nine diverging human mesodermal fates spanning multiple developmental stages (Table S3). This mesoderm gene expression atlas provided insight into the potential cell-of-origin of various human congenital malformations.

Congenital scoliosis is a genetically heterogeneous disease that has been mapped to diverse genes in human patients, in which it causes malformations in the spine, scapulae, and/or ribs (reviewed by Giampietro, 2012; Pourquié, 2011). Due to the inaccessibility of early human embryos, it has been difficult to assess when and where scoliosis susceptibility genes are expressed during development to uncover the origins of this disease.

RNA-seq analysis implicated at least two independent cells-of-origin for different subtypes of congenital scoliosis. For spondylocostal dysostosis (mapped to *DLL3*, *HES7*, *TBX6*, and *LFNG*) and sacral agenesis with vertebral anomalies (*BRACHYURY*), their causative genes were largely expressed in PSC-derived human paraxial mesoderm, but not other mesodermal cell types (Figures 7B and 7Ci). This implies a paraxial mesoderm cell-of-origin for these two types of scoliosis. By contrast, for six other types of congenital scoliosis, their causative genes were strongly expressed in human sclerotome, but not paraxial mesoderm (Figure 7B). Hence, congenital scoliosis may have at least two independent cells-of-origin (paraxial mesoderm or sclerotome) depending on the specific genetic lesion.

RNA-seq analyses of cells diverging across multiple lineage branchpoints also provided a clear view of how fates segregate across consecutive developmental steps. After PS formation, there was a clear partitioning of gene expression patterns in paraxial mesoderm (*TBX6*) and lateral/cardiac mesoderm (*GARP/LRRC32*) (Figure 7Cii). Downstream of paraxial mesoderm, *PAX1* and *PAX3* (Fan and Tessier-Lavigne, 1994) were respectively restricted to either sclerotome or dermomyotome (Figure 7Ciii). These lineage-specific expression patterns demarcate differences in developmental fate and lend confidence to our transcriptional dataset.

The human mesoderm gene expression atlas also uncovered lineage-specific long noncoding RNAs (lncRNAs), nominating them for further study of lncRNA function. By way of example, *Fendrr* is an lncRNA critical for mouse heart field development (Grote et al., 2013), and human *FENDRR* was likewise expressed in hESC-derived cardiac mesoderm (Figure 7Biv).

Mesodermal Distal Regulatory Elements Reflect the Impact of Dynamic Signaling and *trans*-Regulatory Influences on Chromatin

To track how chromatin is dynamically remodeled during development of hESCs into nine distinct types of mesodermal progeny, we charted open chromatin using ATAC-seq to identify putative regulatory elements. We also inferred active TFs for each mesodermal lineage by cross-referencing TFs that were expressed according to RNA-seq with those whose motifs enriched in lineage-specific open chromatin regions (Supplemental Experimental Procedures).

Though accessible chromatin in pluripotent cells was enriched for OCT/SOX-binding motifs, upon 24 hr of differentiation, the PS chromatin landscape became dominated by motifs of TCF/LEF TFs (the effectors of WNT signaling; Figure 7D), reflecting how WNT drives PS induction (Figure 1B). As differentiation progressed, the *trans*-regulatory landscape of day 2 paraxial mesoderm seemed to be built on that of day 1 PS, consistent with how both lineages experience WNT activation (Figure 2): TCF/LEF continued to engage paraxial mesoderm chromatin but was apparently joined by paraxial mesoderm-specific TFs MSGN1 ($p < 10^{-382}$) and TBX6 ($p < 10^{-35}$) (Figure 7D). However, within 24 hr of WNT inhibition and the segmentation of paraxial mesoderm into somites, the landscape transitioned from a TCF/LEF-dominated state to one significantly enriched for FOX motifs ($p < 10^{-61}$; Figure 7D). This was evidenced by predicted FOX motifs in two upstream *MEOX1* enhancers that were accessible in somites (Figure 7F). By virtue of RNA-seq expression patterns (Figure 7E), multiple FOX TFs could account for the FOX-driven somite regulatory state, including FOXC2, which was indeed expressed at the protein level in somites (Figure 2).

Along the alternate lineage pathway, the 24 hr progression from PS to lateral mesoderm involved a transition from a TCF-driven chromatin landscape to a GATA-dominated one (Figure 7D). This reflects the importance of WNT repression in lateral mesoderm specification (Figure 1E) and expression of multiple *GATA* TFs in lateral mesoderm (Figure S7K). Upon 24 hr of further differentiation into GARP$^+$ cardiac mesoderm, GATA motifs became accompanied by HAND1 ($p < 10^{-149}$) and NKX2.5 ($p < 10^{-39}$) (Figure 7D). These findings provide insight into control of mesoderm development: there is no monolithic "pan-mesodermal" program, but instead, chromatin is substantially

Figure 7. The Landscape of Mesoderm Development

(A) Lineage steps with circled numbers correspond to respective sections in the main text and Figure 1A.

(B) RNA-seq expression of human congenital scoliosis genes.

(C) RNA-seq profiling; color intensity depicts gene expression ($\log_2$ TPM) normalized to the expression of that gene in all populations profiled, with the highest-expressing lineage assigned the most intense color value.

(D) ATAC-seq heatmap. Each horizontal line depicts a single chromatin element (left, non-binarized in Figure S7J), with motifs representative of the top four lineage-enriched motifs shown (right).

(E) Inferred *trans*-regulatory lineages programs (left); heatmap of the four FOX TFs most highly expressed in hESC-derived somites (RNA-seq; right).

(F) ATAC-seq of the *MEOX1* locus, with FOX motifs centered in two somitic enhancer elements shown.

See also Figure S7.

remodeled every 24 hr even as closely related mesodermal lineages segue into one another (Figure 7D). Furthermore, the distal element landscape reflects how changes in signaling influences and *trans*-acting regulatory states become physically imprinted on chromatin. Together, this sketches a model for how regulatory states change during mesoderm subtype diversification (Figure 7E).

DISCUSSION

A Roadmap for Human Mesoderm Development

In vitro stem-cell differentiation often yields admixed lineages, possibly due to incomplete suppression of alternate fates or passage through incorrect lineage intermediates. To systematically block production of unwanted lineages in preference to desired fates during stem-cell differentiation, we must map the underlying developmental landscape.

To this end, here we chart a roadmap for human mesoderm development and describe how 12 different human cell types including bone, muscle, and heart emerge from pluripotent cells (Figure 7A). We used scRNA-seq to systematically catalog the diversity of intermediate cell states formed during differentiation, and we tested the minimal combinations of *positive* and *negative* signals that were sufficient for differentiation between each of these intermediate states. Though vertebrate mesoderm development was broadly outlined by pioneering analyses in model organisms that identified certain key genes and signaling pathways (reviewed by Kimelman, 2006; Schier and Talbot, 2005; Tam and Loebel, 2007), it has been difficult to precisely map mesoderm formation due to the large number of mesodermal subtypes and the finely graded, temporally dynamic transitions between them. Throughout consecutive pairwise lineage branches in human mesoderm development, we clearly defined (1) the diverging cell states through scRNA-seq, (2) positive and negative signals inducing each of the mutually exclusive lineages, (3) specific cell-surface markers that identified key mesoderm intermediates, and (4) the chromatin landscapes of the diverging fates (Figure 7A). Besides providing a broad reference map for developmental biology and regenerative medicine, we directly demonstrate the applications of this roadmap to produce engraftable human tissue progenitors and provide insight into developmental signaling dynamics, chromatin remodeling, and congenital disease.

Extrinsic Signals: Logically Blocking Alternative Lineage Formation to Guide Stem-Cell Differentiation

Vertebrate embryology has identified certain signals required for mesoderm formation in model organisms (reviewed by Kimelman, 2006; Schier and Talbot, 2005; Tam and Loebel, 2007), and here, we have tested whether we understand mesoderm development at the level of causation by reconstituting aspects of this process from cultured stem cells. At each lineage transition from human pluripotency to terminally differentiated mesoderm fates, we could identify and test the minimal signaling conditions needed to induce each lineage. The resultant understanding of the underlying signaling logic guided the rapid differentiation of hPSCs into desired mesoderm intermediates (>98% pure MIXL1$^+$ primitive streak; >90% pure NKX2.5$^+$ cardiac mesoderm; >90% pure DLL1$^+$ paraxial mesoderm; >95% pure FOXC2$^+$ early somite progenitors) within several days of differentiation in serum-free, monolayer conditions (without recourse to gene modification). Such efficient, rapid induction relied on the following two principles.

First, the principal finding of this work is that, at each lineage bifurcation, stem cells could be exclusively differentiated down a single lineage path by providing the *positive* signal(s) to induce a given fate while repressing *inhibitory* signal(s) that instead induced the alternate fate. Blocking formation of undesired fates was imperative for efficient differentiation. By way of example, efficient differentiation of day 1 PS into day 2 paraxial mesoderm required WNT activation (to specify paraxial mesoderm) together with inhibition of BMP and TGFβ pathways (to block lateral mesoderm and endoderm formation, respectively) in order to block differentiation toward unwanted fates and to consolidate differentiation down only a single path. Therefore, hPSC differentiation to a desired lineage cannot solely rely on knowledge of the requisite inductive signal(s), but also on an understanding of signals that induce mutually exclusive fates at each step of the way. This underscores the need for systematic developmental roadmaps.

Second, another highlight was the rapidity with which developmental signals were re-interpreted during hPSC differentiation and the consequent importance of controlling temporal signaling dynamics. In the gastrulating mouse embryo, lineage transitions occur every 12–24 hr, for example: E5.5 epiblast → E6.5 primitive streak → E7–7.5 paraxial mesoderm → E8 early somites → E8.5 sclerotome. In vitro, we found that WNT activation on day 0–1 drove hPSC toward PS; WNT activation on day 1–2 then specified paraxial mesoderm; WNT inhibition on day 2–3 differentiated paraxial mesoderm into early somites; and finally, WNT activation on day 3–4 specified dermomyotome. Thus, over the course of 4 days in vitro, WNT was interpreted four different ways as lineages segued into one another every 24 hr. By contrast, some differentiation methods continuously provide the same signal for days or weeks, potentially explaining why a mélange of lineages is produced. Our system therefore constitutes a venue to understand how extrinsic signals are dynamically interpreted in the context of changing windows of developmental competence.

Single-Cell RNA-Seq: Cataloging Mesodermal Lineages and Transition States in between Them

Mapping a developmental hierarchy hinges on cataloging its constituent progenitor states, which we have done here for the human mesodermal lineage hierarchy. We proposed that scRNA-seq sampling (Table S4) would be a complete method to test the lineage and homogeneity of cells at each developmental step. Indeed, early human primitive streak, lateral mesoderm, and paraxial mesoderm lineages were highly uniform. Starting from human paraxial mesoderm, cells initiated somitogenesis along a single continuous trajectory (Figure 2G), and snapshots of this process uncovered the formation of a transient *HOPX*$^+$ human somitomere intermediate that arose for several hours during differentiation. This argues that human somite development entails passage through an ephemeral segmentation-like state (as described for other vertebrate model organisms; Pourquié, 2011), which has been impossible to assess in vivo due to its transient nature and the unavailability of early human embryos.

Navigating Mesoderm Development

Identifying lineage-specific cell-surface markers for major mesoderm subtypes (e.g., DLL1 for paraxial mesoderm and GARP for cardiac mesoderm) enables the purification of desired mesoderm subtypes to investigate the biological characteristics of these cells (as embodied by our RNA-seq and ATAC-seq analyses) or for therapeutic transplantation in the future. Moreover, the major surface markers defined here for human mesoderm progenitors were correspondingly expressed in zebrafish embryos, indicating that they are conserved developmental markers.

Collectively, we delineate a clear lineage hierarchy for mesoderm development with prospectively isolatable lineage intermediates at each step, which should be key for understanding human mesoderm development as well as the clinical purification of hPSC-derived tissue stem and progenitor cells for regenerative medicine in the future (Figure 7A). The ability to produce highly homogeneous populations of human mesodermal progenitors now opens the door to rapid generation and purification of a wealth of different mesodermal cell types from hPSCs—including the engraftable sclerotome and cardiomyocyte populations described here—providing a future foundation for regenerative medicine. Artificially reconstituting aspects of mesoderm development from hPSCs should provide a facile system to study basic developmental processes in vitro, including how developmental signals are temporally re-interpreted and combinatorially integrated and how chromatin dynamics are linked to changing windows of developmental competence.

Yet the roadmap remains incomplete. It does not include lineage paths to human axial mesoderm, intermediate mesoderm, or mediolateral derivatives of sclerotome and dermomyotome (Christ and Scaal, 2008). Finally, though we have identified extracellular signals that specify human mesoderm cell fate in vitro, to accompany the jump of complexity from 2D culture dish to 3D embryo we must in turn map the niche cells that produce these signals during embryogenesis and where they are located, thus unraveling differentiation in 3D space.

EXPERIMENTAL PROCEDURES

Mesoderm Differentiation

Monolayer, feeder-free differentiation was conducted in serum-free CDM2 basal medium (Supplemental Experimental Procedures). hPSCs (mainly H7) were passaged ~1:12–1:20 as fine clumps (using Accutase) onto Geltrex-coated wells and cultured overnight in mTeSR1 + 1 μM thiazovivin. The next morning, hPSCs were differentiated toward anterior PS (30 ng/mL Activin + 4 μM CHIR + 20 ng/mL FGF2 + 100 nM PIK90; for downstream paraxial differentiation) or mid PS (30 ng/mL Activin + 40 ng/mL BMP4 + 6 μM CHIR + 20 ng/mL FGF2 + 100 nM; for downstream lateral differentiation) for 24 hr. Day 1 anterior PS were differentiated toward day 2 paraxial mesoderm (1 μM A8301 + 3 μM CHIR + 250 nM LDN193189 [DM3189] + 20 ng/mL FGF2) for 24 hr. Day 1 mid PS were differentiated toward day 2 lateral mesoderm (1 μM A8301 + 30 ng/mL BMP4 + 1 μM C59) for 24 hr.

Day 2 paraxial mesoderm were differentiated toward day 3 early somites (1 μM A8301 + 1 μM C59 [or alternately, 1 μM XAV939] + 250 nM LDN193189 + 500 nM PD0325901; 24 hr). Day 3 early somites were differentiated toward either day 5–6 sclerotome (5 nM 21K + 1 μM C59) or day 5 dermomyotome (3 μM CHIR + 150 nM Vismodegib, optionally with 50 ng/mL BMP4) for 48–72 hr. Day 5 sclerotome were differentiated toward day 8 fibroblast-like cells (10 ng/mL TGFβ1 + 2 ng/mL PDGF-BB; Cheung et al., 2012) for 72 hr. Day 6 sclerotome were differentiated toward day 9–day 12 cartilage (20 ng/mL BMP4) for 3–6 days.

Ectopic Human Bone Formation

~1.5 × 10^7 day 6 sclerotome cells were subcutaneously transplanted in 1:1 CDM2/Matrigel mixture into NOD-SCID *Il2rg*$^{-/-}$ mice, which were sacrificed in ~2–3 months.

Human Fetal Heart Graft Construction

2 × 7mm strips of week 15–17 human fetal heart were subcutaneously implanted into the ear of NOD-SCID *Il2rg*$^{-/-}$ mice using a trocar. 1 month later, 1.5–2 × 10^6 day 3 cardiac mesoderm or day 8 cardiomyocytes were directly injected into the fetal heart tissue (in 1:1 CDM2/Matrigel mixture) and analyzed ~2.5 months later.

RNA-Seq

RNA was purified from H7-derived mesoderm lineages, either from bulk populations or from 651 single cells spanning ten lineages (Fluidigm C1 system). RNA-seq libraries were prepared (bulk; Ovation RNA-seq System V2 and NEBNext Ultra DNA Library Prep Kit and single cell; SMARTer Ultra Low RNA Kit) and sequenced (Next-Seq 500) to obtain 150 bp paired-end reads, which were processed using the ENCODE long RNA analysis pipeline (Supplemental Experimental Procedures). Collated data is viewable at http://cs.stanford.edu/~zhenghao/mesoderm_gene_atlas.

ACCESSION NUMBER

The accession number for the RNA-seq and ATAC-seq data reported in this paper is SRA: SRP073808.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, seven figures, and seven tables and can be found with this article online at http://dx.doi.org/10.1016/j.cell.2016.06.011.

A video abstract is available at http://dx.doi.org/10.1016/j.cell.2016.06.011#mmc6.

AUTHOR CONTRIBUTIONS

K.M.L., A.C., T.Z.D., J.M.T., N.S.-C., N.B.F., B.M.G., R.E.A.S., and L.T.A. differentiated and transplanted hPSC; P.W.K., R.S., A.A.B., R.M.M., Z.C., and A.K. executed RNA-seq and ATAC-seq; K.Y.S. and W.S.T. stained zebrafish; R.J. and J.A.E. conducted Hopx staining; G.W. and H.V. analyzed heart grafts; and L.T.A., K.M.L., A.C., P.A.B., and I.L.W. oversaw the project.

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